Benchmark Kohn–Sham DFT Calculations on Small Reference Molecules: Orbital Energies, Ionisation Energies, Excitations and Solvation Effects in Preparation for a Computational Study of Organic Photocatalysts

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Abstract

A set of benchmark calculations was carried out to establish the reliability and limitations of the level of theory used in a computational survey of 16 organic photocatalysts, B3LYP/def2-SVP with the ddCOSMO implicit water model in PySCF. Reference molecules with well-characterised properties (water, benzene, anthraquinone, phenol and phenolate) were used to examine (i) the sensitivity of total and orbital energies to basis set and functional, (ii) the quality of optimised geometries, (iii) the physical meaning of Kohn–Sham frontier-orbital energies, (iv) the construction of DFT input from SMILES strings, and (v) the effect of molecular charge and implicit solvation. Optimised geometries agree with experiment to within 0.01 Å. The B3LYP HOMO energy of benzene underestimates the experimental ionisation energy (9.24 eV) by 2.27 eV, whereas the ΔSCF ionisation energy reproduces it to within 0.01 eV. TDDFT places the lowest singlet excitation of benzene at 5.48 eV with zero oscillator strength, consistent with the weak, symmetry-forbidden experimental band at 4.9 eV. The Kohn–Sham gap (6.72 eV) does not correspond to an observable excitation. Implicit water stabilises the HOMO of phenolate by 4.47 eV but that of neutral phenol by only 0.08 eV. Finally, an independent single-point calculation reproduced the photocatalyst workflow's frontier-orbital energies for anthraquinone to four decimal places. These results define how the frontier-orbital data of the photocatalyst study can and cannot be interpreted, and identify TDDFT as the necessary next step.

1 Introduction

The parent study computes the electronic structure of 16 organic photocatalysts drawn from the photoredox literature [1]. They include xanthene dyes, an acridinium, cyanoarenes, aryl ketones, a flavin, a rylene diimide and a porphyrin. The aim is to assess how well a standard density functional approximation reproduces their measured optical and redox properties. The set contains neutral molecules, cations and dianions, heavy-atom substituents, and excitations of local and charge-transfer character. Interpreting the results therefore requires a clear understanding of what the chosen level of theory does well and where it fails.

Before analysing the photocatalysts, the method was tested on small molecules whose properties are known. Five questions were addressed:

1. How strongly do total energies and orbital energies depend on the basis set and functional?
2. How accurate are geometries optimised at this level of theory?

3. What physical quantity, if any, do Kohn–Sham HOMO and LUMO energies represent, and how do they compare with ionisation and excitation energies?
4. How is a molecular structure taken from a SMILES string to a DFT calculation, and how sensitive are the results to the resulting geometry?
5. How do molecular charge and implicit solvation affect frontier-orbital energies?

A final test checks that an independently written calculation reproduces the photocatalyst workflow's results.

2 Computational Methods

All calculations used PySCF 2.14 [2]. Unless stated otherwise, the B3LYP hybrid functional [3, 4] and the def2-SVP basis set [5] were used, matching the photocatalyst study. Closed-shell species were treated with restricted Kohn–Sham theory (RKS) and open-shell species with unrestricted Kohn–Sham theory (UKS). Additional calculations used Hartree–Fock (HF), the PBE functional, and the STO-3G and def2-TZVP basis sets where noted. Implicit solvation used the domain-decomposition conductor-like screening model (ddCOSMO) [6] with $\varepsilon = 78.36$ for water. Density fitting was applied to the Coulomb and exchange integrals for the larger molecules. Geometry optimisations used geomeTRIC [7] with its default convergence criteria; the optimisations in Section 3.2 included the D3(BJ) dispersion correction.

Starting structures were generated from SMILES strings with RDKit. Hydrogens were added explicitly, 3D coordinates were embedded with ETKDGv3 [8] using a fixed random seed, and the structures were relaxed with the MMFF94 force field [9]. The resulting element symbols and Cartesian coordinates were passed to PySCF, together with the total charge from the SMILES formal charges. Excitation energies and oscillator strengths were computed with linear-response TDDFT. Vertical ionisation energies were obtained in two ways: from Koopmans' theorem [10] and by the ΔSCF method,

$$\mathrm{IE}_{\mathrm{Koopmans}} \approx -\varepsilon_{\mathrm{HOMO}}, \qquad \mathrm{IE}_{\Delta\mathrm{SCF}} = E(\mathrm{M}^{+}) - E(\mathrm{M}), \tag{1}$$

where both total energies are evaluated at the geometry of the neutral molecule M. Energies are reported in eV ($1\,E_\mathrm{h} = 27.2114$ eV) and wavelengths in nm ($\lambda = 1239.84/E$). Scripts were developed with AI coding assistance; all results were checked against the reference values given below.

3 Results and Discussion

3.1 Sensitivity to basis set and functional

Table 1 lists total energies and HOMO energies of water at its experimental geometry. Enlarging the basis from STO-3G to def2-TZVP lowers the total energy by about 1.1 E_h ($\approx$3000 kJ mol^{-1}), as expected from the variational principle. Changing from HF to B3LYP lowers it by about 0.4 E_h. Both changes are far larger than typical reaction or excitation energies. Absolute total energies are therefore meaningful only as differences between calculations at an identical level of theory.

Orbital energies are even more sensitive to the method. With def2-TZVP, the HF HOMO (-13.82 eV) lies close to the negative of the experimental ionisation energy, while the B3LYP HOMO (-8.63 eV) is about 4 eV too high. This difference is examined in Section 3.3.

3.2 Optimised geometries

Table 2 compares optimised and experimental structural parameters. The C–C bond length of benzene is reproduced to within 0.001 Å, and the C–H and O–H bonds are about 0.01 Å too long,

Table 1: Water at the experimental geometry: total energies and HOMO energies as functions of basis set and method. Experimental vertical ionisation energy: 12.6 eV.

		E_{tot} (E_{h})			$\varepsilon_{\mathrm{HOMO}}$ (eV)	
Basis	N_{AO}	HF	PBE	B3LYP	HF	B3LYP
STO-3G	7	−74.963	−75.226	−75.313	−10.65	−3.92
def2-SVP	24	−75.961	−76.272	−76.358	−13.55	−7.94
def2-TZVP	43	−76.059	−76.376	−76.463	−13.82	−8.63

which is typical of DFT with a double-ζ basis. The water bond angle is underestimated by 1.4°. Starting from the MMFF94 structure, the benzene optimisation converged in four cycles with an energy change of only 0.14 kcal mol^{-1}, showing that the force-field geometry was already close to the DFT minimum.

Table 2: Optimised (B3LYP-D3(BJ)/def2-SVP) and experimental geometries.

Molecule	Parameter	Calculated	Experiment	Cycles
H_2O	r(O–H) / Å	0.967	0.958	3
	$\angle$(H–O–H) / °	103.1	104.5	
C_6H_6	r(C–C) / Å	1.398	1.397	4
	r(C–H) / Å	1.093	1.084	

3.3 Frontier orbitals, ionisation energy and excitations of benzene

Orbital energies. The B3LYP/def2-SVP calculation on optimised benzene (42 electrons, 114 basis functions) gives a doubly degenerate HOMO at −6.97 eV and a doubly degenerate LUMO at −0.25 eV. These are the e_{1g} and e_{2u} π orbitals of D_{6h} symmetry, with one and two nodal planes respectively (Figure 1). Because any linear combination of a degenerate pair is an equally valid solution, the orientation of an individual orbital has no physical significance.

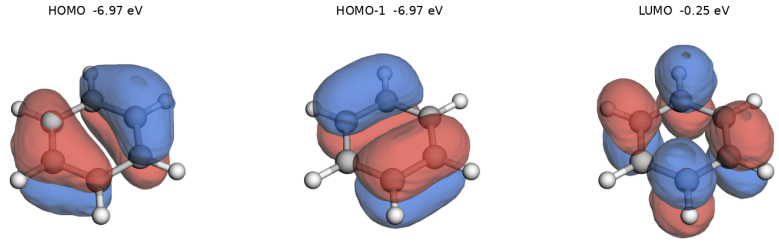


Figure 1: Frontier orbitals of benzene (B3LYP/def2-SVP; isosurfaces at ±0.03 a.u.). HOMO and HOMO−1 form the degenerate e_{1g} pair; the LUMO belongs to the degenerate e_{2u} pair.

Ionisation energy. Table 3 compares the two estimates of Eq. (1) with experiment. The B3LYP Koopmans estimate underestimates the ionisation energy by 2.27 eV, whereas the ΔSCF value agrees with experiment to within 0.01 eV. The discrepancy is due to the self-interaction error of approximate exchange–correlation functionals. The Hartree (Coulomb) term includes each electron's interaction with its own density, and approximate functionals cancel this only partially. The remaining spurious self-repulsion destabilises occupied orbitals and raises $\varepsilon_{\mathrm{HOMO}}$. ΔSCF avoids orbital energies entirely. It uses differences of total energies, in which much of the error cancels, and it includes the relaxation of the remaining electrons after ionisation. The HF Koopmans value (9.17 eV) is also close to experiment. HF has no self-interaction error, and its

two main omissions, orbital relaxation (which makes the IE too large) and electron correlation (which makes it too small), partly cancel.

Table 3: Vertical ionisation energy of benzene (def2-SVP).

Method	IE (eV)	Deviation (eV)
Experiment	9.24	—
$-\varepsilon_{\mathrm{HOMO}}$, B3LYP	6.97	−2.27
$-\varepsilon_{\mathrm{HOMO}}$, HF	9.17	−0.07
ΔSCF, B3LYP (UKS cation)	9.24	0.00

Excitation energies. Table 4 lists the lowest TDDFT excitations. The S_1 and S_2 states have zero oscillator strength: they are symmetry-forbidden at the D_{6h} geometry. This is consistent with the weak experimental band at 254 nm (4.9 eV), which gains intensity only through vibronic coupling. The first allowed transition (degenerate S_3/S_4, $f = 0.59$) lies at 7.28 eV, compared with about 6.9 eV experimentally. The TDDFT errors of 0.4–0.6 eV are somewhat larger than the ≈0.3 eV typical of hybrid functionals over large benchmark sets [11], as expected for a small double-ζ basis. The Kohn–Sham HOMO–LUMO gap (6.72 eV) differs from the lowest excitation by 1.2 eV and contains no information about which transitions are allowed.

Table 4: Lowest TDDFT singlet excitations of benzene (B3LYP/def2-SVP) compared with the Kohn–Sham gap and experiment.

	E (eV)	λ (nm)	f	Experiment
KS gap, $\varepsilon_{\mathrm{LUMO}} - \varepsilon_{\mathrm{HOMO}}$	6.72	184	—	—
S_1	5.48	226	0.000	4.9 eV (254 nm), weak
S_2	6.22	199	0.000	—
S_3, S_4	7.28	170	0.589	≈6.9 eV, strong

These results separate three quantities that are often confused. The *Kohn–Sham gap* is a difference of orbital eigenvalues. The *fundamental gap*, IE − EA, refers to removing and adding an electron in separate systems. The *optical gap* is the lowest neutral excitation, which is lowered relative to the fundamental gap by the electron–hole attraction. Only the optical gap corresponds to an absorption spectrum, and it requires an excited-state method such as TDDFT.

3.4 From SMILES to a DFT calculation

An RDKit molecule constructed from a SMILES string is a molecular graph. It contains atoms, bonds, formal charges and implicit hydrogen counts, but no Cartesian coordinates. DFT, in contrast, requires only nuclear positions, total charge and spin multiplicity, together with the choice of method. Explicit hydrogens, ETKDG embedding and MMFF94 relaxation bridge the two. For benzene this route gave r(C–C) = 1.395 Å, and a HOMO energy within 0.008 eV of that at the DFT-optimised geometry.

Table 5 shows the corresponding results for anthraquinone ($C_{14}H_8O_2$; 108 electrons, 264 basis functions). The geometry (MMFF94 versus DFT-optimised) shifts both frontier orbitals by 0.02–0.03 eV in the same direction, so the gap changes by only 0.007 eV. Implicit water shifts the HOMO by −0.12 eV and the gap by +0.07 eV. These effects are small because anthraquinone is rigid, has a single low-energy conformer, and is neutral with a negligible dipole moment.

A practical point concerns the charge. PySCF can read Cartesian coordinates directly from an xyz file, but the xyz format has no charge field. When the eosin Y dianion's structure file was read this way, the molecule was assigned charge 0 and 306 electrons instead of 308, without any warning. The total charge must therefore always be supplied explicitly.

Table 5: Frontier-orbital energies of anthraquinone (B3LYP/def2-SVP).

Geometry	Solvent	$\varepsilon_{\mathrm{HOMO}}$ (eV)	$\varepsilon_{\mathrm{LUMO}}$ (eV)	Gap (eV)
MMFF94	vacuum	−7.170	−2.976	4.194
MMFF94	water (ddCOSMO)	−7.288	−3.022	4.266
DFT-optimised	water (ddCOSMO)	−7.314	−3.041	4.273

3.5 Effect of charge and implicit solvation

Phenol and its conjugate base, phenolate, were used as minimal models of the deprotonated phenol oxygen found in eosin Y and related xanthene dyes (Table 6). In vacuum the phenolate HOMO is positive (+0.32 eV): the excess electron is unbound at this level of theory. The absence of diffuse basis functions and self-interaction error both contribute to this. Implicit water lowers the phenolate HOMO by 4.47 eV, compared with only 0.08 eV for neutral phenol, because the dielectric continuum responds mainly to the net charge.

Table 6: HOMO energies of phenol and phenolate in vacuum and in implicit water (B3LYP/def2-SVP, MMFF94 geometries).

	$\varepsilon_{\mathrm{HOMO}}$, vacuum (eV)	$\varepsilon_{\mathrm{HOMO}}$, water (eV)	Shift (eV)
Phenol, C_6H_5OH	−6.18	−6.26	−0.08
Phenolate, $C_6H_5O^-$	+0.32	−4.15	−4.47

When phenolate was built with the default charge of zero, the calculation was rejected because 49 electrons cannot form a closed-shell singlet. For a dianion, omitting the charge removes two electrons, leaving an even electron count. The calculation then runs without error on the wrong species, as in the eosin Y example above.

3.6 Reproduction of a photocatalyst-workflow result

A single-point calculation written independently, using the procedure of the preceding sections, was run on the workflow's B3LYP/def2-SVP-optimised anthraquinone geometry in implicit water. It gave $\varepsilon_{\mathrm{HOMO}} = -7.3138$ eV and $\varepsilon_{\mathrm{LUMO}} = -3.0409$ eV, identical to the workflow's stored values to four decimal places. This confirms that the workflow's frontier-orbital step performs the calculation described here.

4 Implications for the Photocatalyst Study

1. **Frontier-orbital energies.** The B3LYP HOMO and LUMO energies of the 16 photocatalysts are shifted by self-interaction error of the order of 2 eV (Table 3). They can be used to compare molecules and identify trends, but not as absolute ionisation energies, electron affinities or redox potentials. Plotting them against the water-splitting redox levels is therefore not justified. Redox potentials should instead be computed as ΔSCF free-energy differences in solvent.
2. **Optical properties.** The study's current comparison of the Kohn–Sham gap with the lowest experimental absorption band measures a trend, not absorption energies. Excitation energies, oscillator strengths and the character of each excitation require TDDFT. A range-separated functional is advisable for the charge-transfer catalysts (Mes-Acr^+, 4CzIPN).
3. **Geometry.** For rigid, neutral molecules, MMFF94 geometries change the frontier orbitals by only about 0.01 eV. Flexible or charged dyes require separate verification.

4. **Charge and solvation.** The dianionic and cationic dyes must carry their charge explicitly and be treated with a solvent model. Using the same implicit solvent for all 16 molecules keeps the level of theory consistent, so they can be compared directly. For neutral, non-polar molecules this has little effect; for strongly polar neutral molecules, such as riboflavin, it may matter more.

5 Conclusions

Benchmark calculations on small reference molecules have established the behaviour of the B3LYP/def2-SVP/ddCOSMO level of theory used in the photocatalyst study. Geometries are accurate to about 0.01 Å. ΔSCF reproduces the ionisation energy of benzene, while the Kohn–Sham HOMO underestimates it by 2.27 eV. TDDFT correctly identifies the lowest singlet excitation of benzene as dark, and it lies 1.2 eV below the Kohn–Sham gap. Implicit solvation is essential for anions (4.47 eV stabilisation of the phenolate HOMO) but has little effect on neutral, non-polar molecules. An independent calculation reproduced the workflow's frontier-orbital energies for anthraquinone exactly. The next steps are to add TDDFT excitation energies to the photocatalyst study, beginning with anthraquinone (experimental absorption at 325 nm), and to verify the frontier orbitals of the flexible dyes at their DFT-optimised geometries.

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